MILITARY SPECIFICATION SKEET
GUIDANCE CONTROL SECTION, GUIDED MISSILE
G-BIAS/NAV GAIN ASSEMBLY

This specification is approved for use with the Naval Air Systems Command., Department of the Navy, and is available for use by all departments and agencies of the Department of defense.

The requirements for acquiring the G-kiss/Nav Gain described herein shall consist of this specification and the latest issue of MIL-G-85742.

1. SCOPE

l.1 Scope. This specification sheet establishes the performance and test requirements for the G-Bias/Nav Gain Assembly used in the Guidance Control Section of the SIDEARM Missile.

2. APPLICABLE DOCUMENTS.

2.1 Government drawings. The following drawings, in addition to the applicable documents listed in MIL-G-85742, are applicable to this specification sheet.

Naval Air Systems Command
(Code Ident 30003)

1466As1324	G-Bias/Nav Gain Assembly
1466AS1326	PW3 Assembly G-Bias

3. REQUIREMENTS

3.1 Item description. The G-Bias/Nav Gain (GB/NG) assembly is a part of the receiver section. provides steering data to the magnetic amplifier (Magamp) during flight in order to guide the missile to its target. The G-Bias function senses the gravity vector direction and outputs signals which are translated into fin torque used to counteract the force of gravity. These signals are combined on the Nav Gain board with steering data from the seeker section. A navigation gain function is superimposed on the steering data in order to vary the magnitude of the fin torque as a function of time.

FSC 1420

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3.2 Characteristics. Performance characteristics of the BG/NG shall be as specified herein.

3.2.1 Rate gyro drive voltage. The rate gyro drive voltage shall be 9.0 ±0.3 volts root-mean-square at a frequency of 1200325 hertz (Hz).

3.2.2 Roll angle drift. The roll angle voltage at the output of the integrator shall drift not more than 15 millivolts (mV) (approximately 1 degree (deg)) in 10 seconds.

3.2.3 Roll angle accuracy. When rotating the assembly about its longtudinal axis at a rate of 60 degrees per second (deg/s), the cumulative angle error after 30 seconds shall be less than 1 percent.

3.2.4 G-Bias output voltages. The G-Bias output voltages shall vary as the unit rolls from its launch position. The output shall conform to Figure 1 for 0 deg initialization, Figure 2 for +90 deg initialization, and Figure 3 for -90 deg initialization. These requirements shall be met for roll rates up to 200 deg/s.

3.2.5 Torque steering data delay. Upon enablement, the G3/NG assembly shall provide a 2.1 ± 0.1 second delay prior to incorporating seeker-steering data into the Magamp Input.

3.2.6 Navigation gain function. The normalized magnitude of the steering data at the Magamp In outputs shall vary in accordance with Figure 4.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. Each GB/NG assembly shall meet the requirements of 3.2 when inspected and tested as specified in accordance with Drawing 1466AS1324 and the following paragraphs. Failure to meet any of the requirements shall be cause for rejection of the GB/NG assembly.

4.1.1 Test environment. Unless otherwise specified, all tests and inspections shall be conducted under the following environmental conditions.

a. Temperature (ambient): 23 ±5 degrees Celsius.

b. Altitude: Elevation of test facility.

c. Relative humidity: 95 percent or less.

4.1.2 Test equipment accuracy. Unless otherwise specified, test equipment accuracy shall be as specified in MIL-G-85742.

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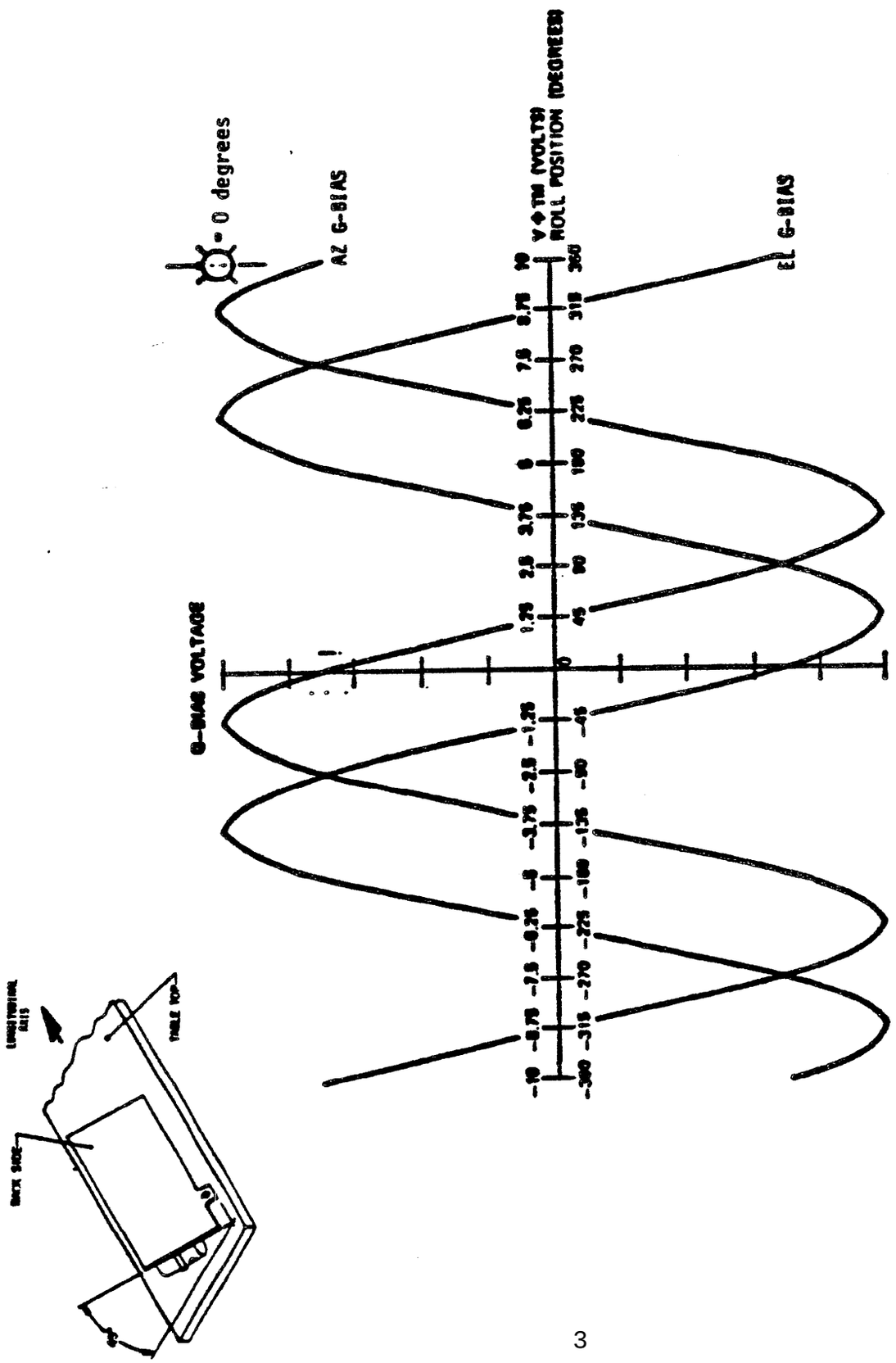


FIGURE 1. 0 bias AZ and EL outputs versus roll position for 0 degrees Initialization.

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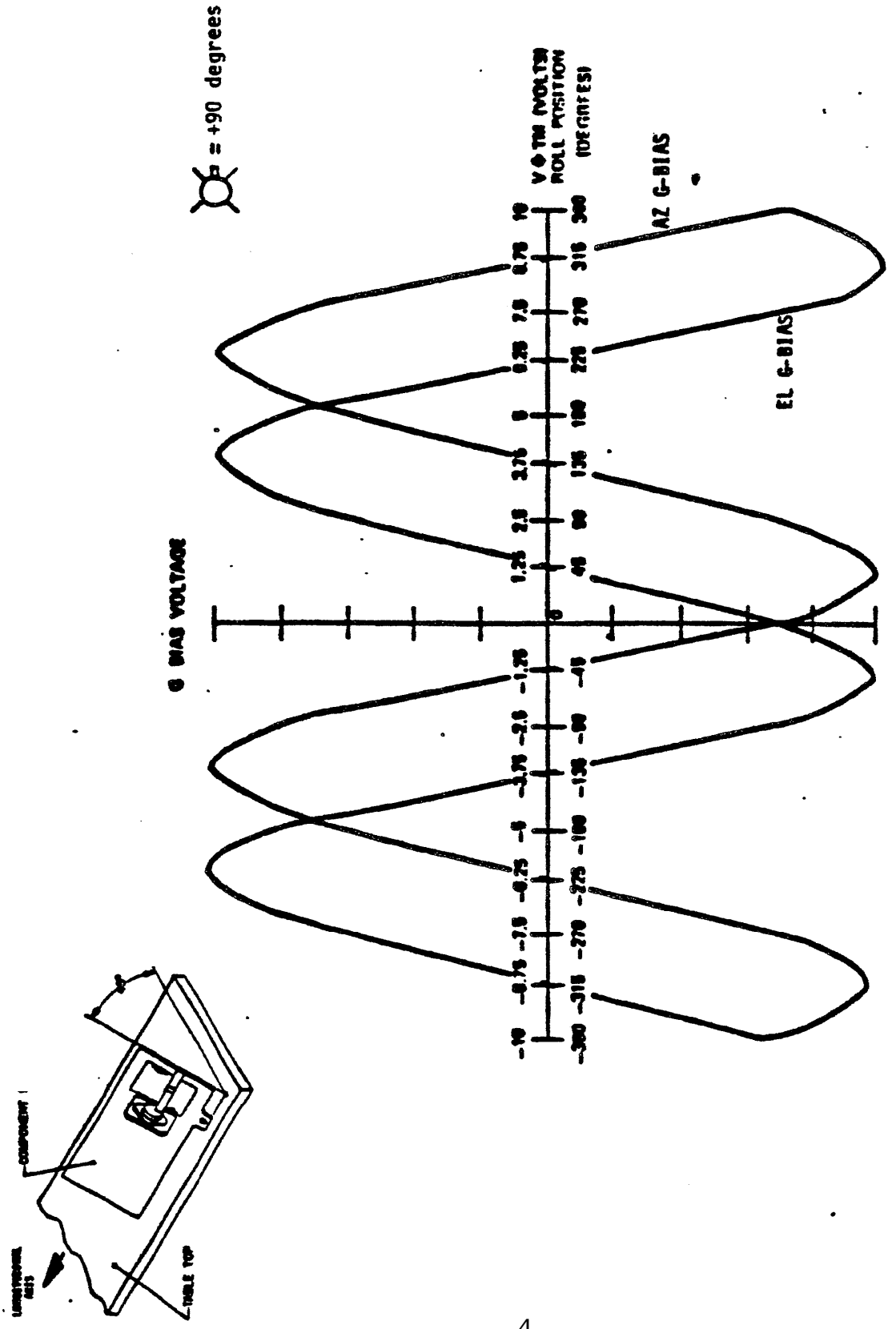


FIGURE 2. G bias AZ and EL outputs versus roll position for +90 degrees initialization.

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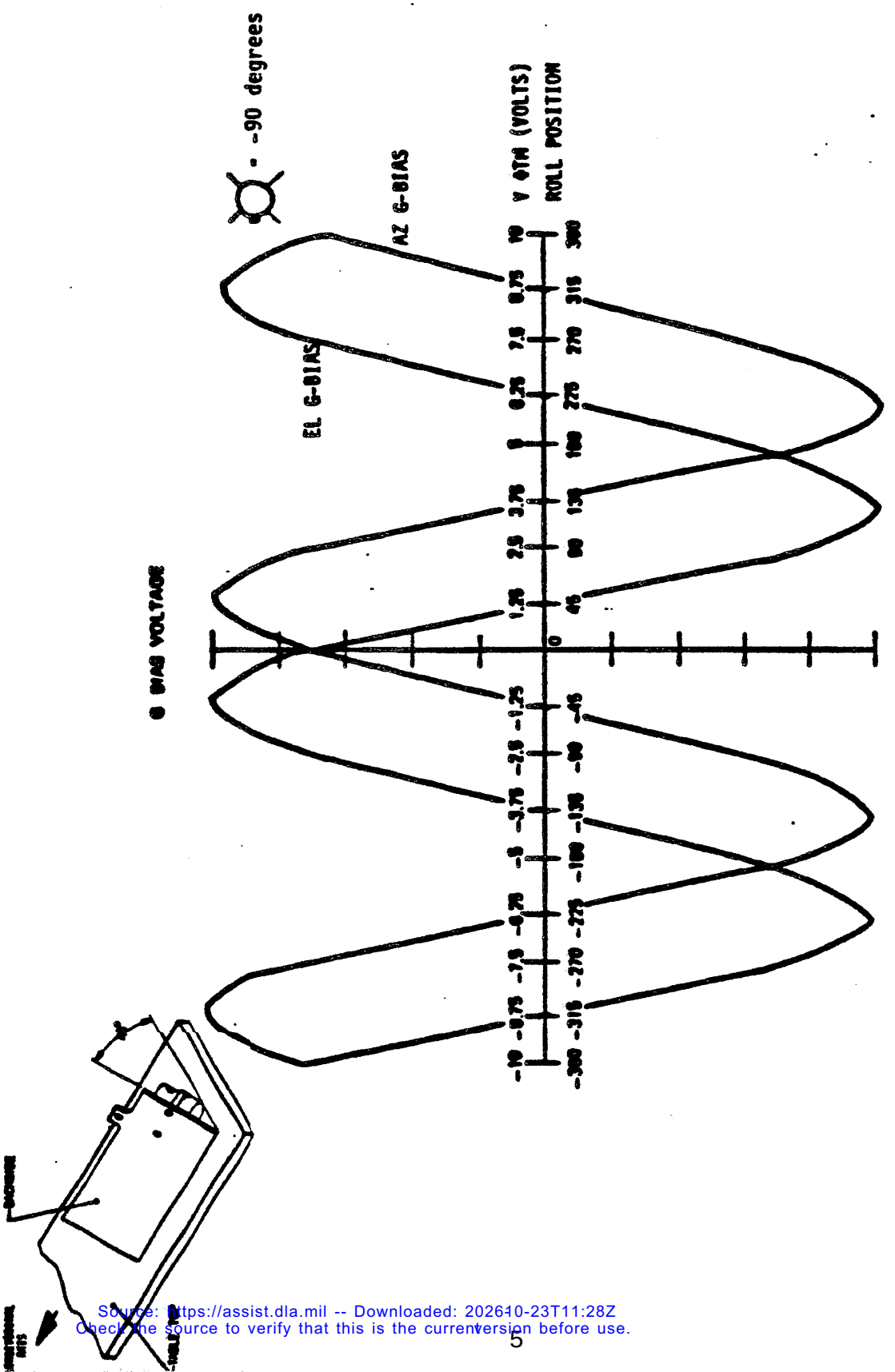
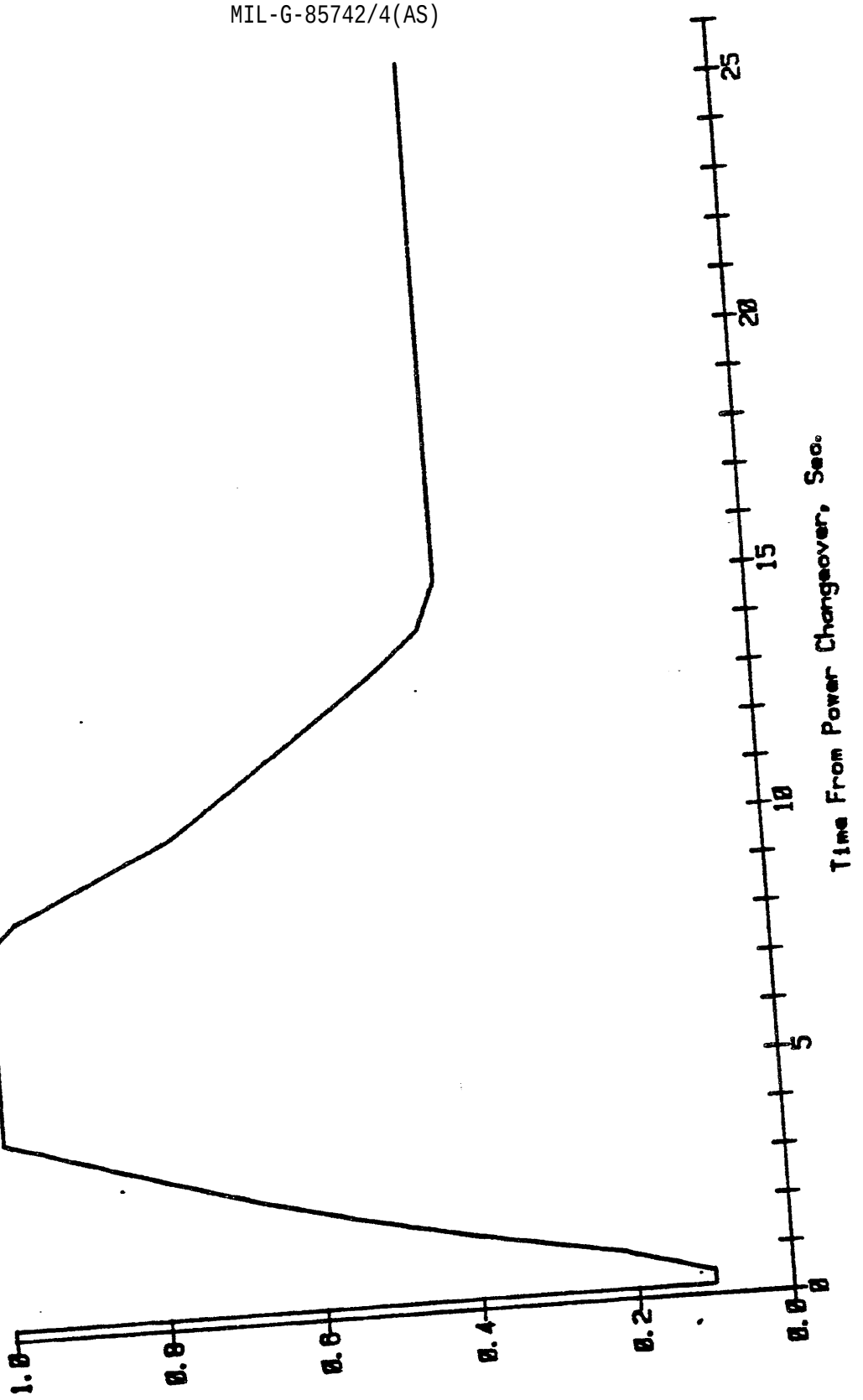


FIGURE 3. G bias AZ and EL outputs versus roll position for -90 degrees initialization.

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4.2 Quality conformance inspections.

4.2.1 Unit tests. The assembly shall be subjected to a unit test in order to establish conformance to the requirements specified herein.

4.2.2 Input voltages. External power supplies providing +15.00 + 0.15 volts V), -15.00 ± 0.15 V, and +5.00 ± 0.15 V shall be applied at pins 11, 12, and 10, respectively, using pin 1 as return (connector A2A2P1).

4.2.3 Test methods. Details of methods and procedures shall be in accordance with the Government approved test procedure.

4.2.3.1 Rate gyro drive voltage. Measure the voltage on E15 of the G-Bias board (Drawing 1466AS1326) and verify that it meets the requirements specified in 3.2.1.

4.2.3.2 Roll angle drift. With the G-Bias inhibit line grounded, measure the voltage on pin 20 of connector A2A2P1 (should be less than 20 mV). Remove the ground from the G-Bias inhibit line. The voltage shall meet the requirements of 3.2.3.

4.2.3.3 Roll angle accuracy. Measure the voltage on pin 20 of connector A2A2P1. Rotate the unit about its longitudinal axis at a rate of 60 deg/s. Compare the periods of the saw-toothed roll angle output to the time of the actual revolutions. Calculate the percent error as follows:

$$\text{Percent error} = 1 - \frac{\text{time of five roll angle periods}}{\text{time of five revolutions}}$$

The error shall meet the requirements of 3.2.4.

4.2.3.4 G-Bias output voltages. Initialize the unit in each of the three positions shown in Figures 1, 2, and 3. Measure the Magamp IN outputs as the unit is rotated about its longitudinal axis at a roll rate of 60 deg/s. The voltage waveforms shall meet the requirements of 3.2.5.

4.2.3.5 Torque steering data delay. Ground the G-Bias inhibit line and apply 10 V to the GB/NG input disable line. Apply 200 mV to the Az and E1 torque telemetry outputs and 4 V to the Az and E1 G-Bias telemetry outputs- Measure the Magamp in outputs. Remove the 10 V from the GB/NG input disable line and verify a discrete voltage step of approximately 1.5 V in both Magamp In signals. After a delay of the amount specified in 3.2.5, a second discrete step of approximately 1.5 V shalt occur.

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4.2.3.6 Navigation gain function. Initialize the unit in the position shown in Figure 1. Ground the Nav Gain Set line and apply 10 V to the GB/NG input disable line. Measure the Magamp In outputs and remove the 10 V from the GB/NG input disable line. The outputs shall meet the requirements of 3.2.6.

Preparing activity
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(Project 1420-NO4)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-85742/4(AS)	G-Bias/NAV Gain Assembly

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: ____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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